Fiscal Year:	FY 2019	**Task Last Updated:**	FY 12/28/2018
PI Name:	Foster, Jamie Ph.D.		
Project Title:	Impact of Microgravity on the Cell-Cell Interactions between a Mutualistic Bacterium and Its Animal Host		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	jfoster@ufl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	321-525-1047
Organization Name:	University of Florida		
PI Address 1:	Department of Microbiology and Cell Science		
PI Address 2:	Space Life Science Laboratory, 505 Odyssey Way		
PI Web Page:			
City:	Merritt Island	**State:**	FL
Zip Code:	32953	**Congressional District:**	3
Comments:	EDITOR'S NOTE: Congressional district is for University of Florida, Gainesville. Address is PI's physical address.		
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	12/01/2013	**End Date:**	11/30/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
Contact Email:	kevin.y.sato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2018 per F. Hernandez/ARC (Ed., 11/14/17) NOTE: End date changed to 11/30/2017 per NSSC information (Ed., 9/6/16)		
Key Personnel Changes/Previous PI:	March 2017 - The postdoctoral fellow working on this project received a job and left this project. However, in May 2017 a new graduate student joined the project and will use this research as part of her dissertation research.		
COI Name (Institution):			
Grant/Contract No.:	NNX13AM44G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Beneficial microbes play a significant role in the normal development of animal tissues. Through the formation of healthy stable associations between host and symbiont, particularly during post-embryonic development, carefully regulated developmental pathways can be induced. In this experimental research plan, we propose to examine the impact of microgravity on the normal molecular interactions between host and symbiont. To examine these effects we propose to use the model system between the squid Euprymna scolopes and the luminescent bacterium Vibrio fischeri. For over two decades this squid-vibrio model has served as an ideal ecosystem to examine the influences of mutualistic bacteria on animal development. To simulate microgravity, rotating bioreactors called high-aspect-ratio rotating wall vessel bioreactors (HARVs) will be used to provide a low-shear environment for both the host and symbiont, thereby simulating microgravity. Specifically, we propose the following: Objective 1. Examine the mechanism(s) associated with the delay of bacteria-induced immune cell trafficking that occurs in simulated microgravity. Objective 2. Characterize the changes in the transcriptome of the host squid in the presence and absence of symbiotic bacteria exposure under microgravity conditions. Objective 3. Develop a flight-ready microtiter plate for future flight experiments and optimize the conditions needed to initiate and maintain the symbiosis in this new hardware. The proposed work has several implications for the field of space biology. First, by examining how normal healthy symbioses are initiated and maintained under microgravity conditions, we can begin to determine the sensitivity of these ecosystems to perturbations in the space environment. Secondly, we can assess the genetic and metabolic responses to gravity in the presence and absence of symbiosis, potentially identifying previously unknown triggers or cues associated with post-embryonic animal development.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	All animals form symbiotic associations with bacteria. These ubiquitous interactions are essential for the normal development and health of the host organism. This project helps to improve our understanding of the molecular cross-talk that occurs between animals and microbes by delineating those genes and pathways that are influenced by gravity. As the force of gravity has not changed throughout Earth's history we have a unique opportunity to assess how these host-microbe interactions alter gene expression and their physiology in the absence of gravity.
Task Progress:	A. Project Overview – This project examined the impact of microgravity on commensal, non-pathogenic microbes that associate with animal tissues. To address this issue we conducted an integrative study in which the responses of both animal host and microbe were examined in situ under simulated microgravity conditions. The long-term goal of the research was to assess whether those interactions between animal cells and mutualistic bacteria undergo a significant alteration in cell-cell communication, immune response, and physiology under microgravity conditions. The overall objective of this proposal was to examine the genomic expression patterns of a host symbiotic squid Euprymna scolopes in response to the luminescent bacterial symbiont, Vibrio fischeri. The results of this study helped identify those genetic pathways that are differentially regulated in the host animal in the presence and absence of mutualistic bacteria under simulated microgravity conditions, thereby improving our understanding of human health in the space environment. B. Major Findings and Outcomes of Research 1. The symbiotic V. fischeri appear to modulate the host innate immune and oxidative stress response of the host light organ during modeled microgravity conditions. Using RNA-seq we analyzed the transcriptome of the host light organ (i.e., the site of symbiosis) of the squid in the presence (symbiotic) and absence (aposymbiotic) of beneficial V. fischeri under simulated microgravity conditions. At 12 h post inoculation both the symbiotic and aposymbiotic animals exhibited an upregulation of genes associated with hemocyanin, oxidative stress, and the innate immune system. However, by 24 h post inoculation only the aposymbiotic animals exhibited an upregulated stress response, whereas those animals exposed to V. fischeri no longer showed the stress response suggesting that V. fischeri is helping to modulate the host response. Results were published in Casaburi et al., 2017 – See cumulative bibliography for citation; reported in FY2018 report. 2. Lipopolysaccharide (LPS) shedding by V. fischeri is increased under simulated microgravity conditions. Under modeled microgravity conditions V. fischeri releases a statistically higher amount of the virulence factor LPS, known to cause apoptosis and inflammation in both mutualistic and pathogenic microbes. The LPS production is primarily associated with flagellar movement and by knocking out flagellar production and motion the amount of LPS produced under both microgravity and gravity conditions was reduced. However, even with the removal of the flagella, the LPS shedding was still statistically higher in modeled microgravity conditions compared to gravity conditions. We have ascertained that under modeled microgravity the V. fischeri membrane integrity is compromised and is more susceptible to antibiotic treatment. We are examining whether we can further inhibit the differential shedding of LPS under modeled microgravity conditions. 3. Sequencing and closure of the Euprymna scolopes genome revealed two evolutionary mechanisms for symbiotic organ evolution. Using the RNA-seq data generated during this project we generated a reference transcriptome for the host animal and use that as a scaffold to close the genome of E. scolopes. Although this task was not originally part of our project goals the data was invaluable in providing a closed genome for the squid-vibrio research community. Analysis of the genome and reference transcriptome revealed that there are two distinctive evolutionary processes associated with the origin of symbiotic organs in animals. A manuscript is currently in press at PNAS. 4. Modeled microgravity alters the innate immune pathway of host squid. Using the new technique NanoString, we complemented the previously generated transcriptomes by analyzing the differential gene expression along additional time points (0, 2, 6, 8, 10, 12, 14, 16, 18, 24 h) and have generated a new model of the receptor and signal cascade of the innate immune system. We have also determined that modeled microgravity impacts the expression of the innate immune genes and we have identified key parts of the pathway that are being disrupted. (Publication in preparation.) 5. Modeled microgravity does not significantly alter the transcriptome of the symbiont V. fischeri; however, the global regulatory gene hfq is critical to regulate the bacterial stress responses in low shear modeled microgravity (LSMMG). The transcriptome of the bacterial symbiont was examined during both exponential and stationary phase growth and

	results indicated that modeled microgravity had no significant impact on gene expression at 12 and 24 h. However, when the global regulator gene hfq was deleted from the bacteria, there was a pronounced change in gene expression under LSMMG conditions. There was a dramatic increase in the number of transcripts associated with TCA cycling and stress responses suggesting that the Hfq protein serves as a critical regulator for the bacteria under modeled microgravity conditions. (Published Duscher et al., 2018.)
Bibliography Type:	Description: (Last Updated: 04/20/2026)
Articles in Peer-reviewed Journals	Belcaid M, Casaburi G, McAnulty SJ, Schmidbaur H, Suria AM, Moriano-Gutierrez S, Pankey MS, Oakley TH, Kremer N, Koch EJ, Colins AJ, Nguyen H, Lek S, Goncharenko I, Minx P, Sodergren E, Weinstock G, Rokhsar D, McFall-Ngai MJ, Simakov O, Foster JS, Nyholm SV. "Symbiotic organs shaped by distinct modes of genome evolution in cephalopods." Proceedings of the National Academy of Sciences. 2019 Feb 19;116(8):3030-5. https://doi.org/10.1073/pnas.1817322116 ; PubMed PMID: 30635418; PubMed Central PMCID: PMC6386654 , Feb-2019
Articles in Peer-reviewed Journals	Duscher AA, Conesa A, Bishop M, Vroom MM, Zubizarreta SD, Foster JS. "Transcriptional profiling of the mutualistic bacterium Vibrio fischeri and an hfq mutant under modeled microgravity." npj Microgravity. 2018 Dec 18;4(1):25. eCollection 2018. eCollection 2018. https://doi.org/10.1038/s41526-018-0060-1 ; PubMed PMID: 30588486; PubMed Central PMCID: PMC6299092 , Dec-2018
Articles in Peer-reviewed Journals	Brooks JF, Gyllborg MC, Kocher AA, Markey LE, Mandel MJ. "TfoX-based genetic mapping identifies Vibrio fischeri strain-level differences and reveals a common lineage of laboratory strains." Journal of bacteriology. 2015 Mar 15;197(6):1065-74. https://doi.org/10.1128/jb.02347-14 , Mar-2015
Journal/Magazine covers	Belcaid M, Casaburi G, McAnulty SJ, Schmidbaur H, Suria AM, Moriano-Gutierrez S, Pankey MS, Oakley TH, Kremer N, Koch EJ, Colins AJ, Nguyen H, Lek S, Goncharenko I, Minx P, Sodergren E, Weinstock G, Rokhsar D, McFall-Ngai MJ, Simakov O, Foster JS, Nyholm SV. "Cover in journal Proceedings of the National Academy of Sciences USA for article, 'Symbiotic organs shaped by distinct modes of genome evolution in cephalopods.' " Proc Natl Acad Sci U S A. 2019 Feb 19;116(8):3030-5. https://doi.org/10.1073/pnas.1817322116 ; PubMed PMID: 30635418; https://www.pnas.org/content/116/8.cover-expansion , Feb-2019